

Problem 1

By Lagrange's Theorem we know that S_4 has subgroups of order 1, 2, 3, 4, 6, 8, 12

Let H be a subgroup of order 12. By Cauchy's Theorem we know it has a subgroup of order 3. Thus, it has a 3 cycle. Since it has a 3 cycle we know it is A_4 since A_4 is generated by 3 cycles, and this is normal since it has index 2.

Let H be a group of order 8. These must be 2-Sylow subgroups and by Sylow 3 we know there is 1 or 3 but there is 3 since two examples are $\langle (1234), (13) \rangle$ and $\langle (1243), (14) \rangle$ Thus these are all conjugate by Sylow 2 thus these are not normal.

Let H be a group of order 6. This must be S_3 since there is no elements of order 6 in S_4 . This means it has a 3 cycle but since all 3 cycles are conjugate we know that we can get an element outside the subgroup since the S_3 subgroup will only have 2 3-cycles.

Let H be a group of order 4. This is either C_4 or $C_2 \times C_2$. If it is C_4 then it is a cycle group generated by a 4 cycle, and since $\sigma(abcd)\sigma^{-1} = (\sigma(a)\sigma(b)\sigma(c)\sigma(d))$ we know that all 4 cycles are conjugate. Thus, since there 6 4-cycles we know we can get outside our group of 4.

If it is $C_2 \times C_2$ then if it has a transposition then we can get out again by similar proof by showing all transpositions are conjugate. Thus the only possibility is $\langle (12)(34), (13)(24) \rangle$

Since these are a product of disjoint cycles if we conjugate $\sigma(ab)(cd)\sigma^{-1} = (\sigma(a)\sigma(b))(\sigma(c)\sigma(d))$ then since σ permutes elements we know that this will also result in a product of disjoint transpositions and we have them all in this subgroup this is normal.

Let H be a group of order 3. Similar to order 4 for C_4 this is not normal

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Thus we only have normal groups $\langle (12)(34), (13)(24) \rangle$, A_4 and $\{e\}$

For normal in A_4 we know from previous homework that it has normal $C_2 \times C_2$ and this could only be $\langle (12)(34), (13)(24) \rangle$

Thus all composition series are

$$S_4 \supseteq A_4 \supseteq \langle (12)(34), (13)(24) \rangle \supseteq \{1, (12)(34)\} \supseteq \{e\}$$

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Problem 2

- (a) For A_4 , Consider the homomorphism $\varphi : A_4 \rightarrow A_4 / \langle (12)(34), (13)(24) \rangle$ since $|\langle (12)(34), (13)(24) \rangle| = 4$ and is normal we know $A_4 / \langle (12)(34), (13)(24) \rangle \cong C_3$ which is abelian.
 Consider $[a, b] \in A'_4$, $aba^{-1}b^{-1}$. $\varphi(aba^{-1}b^{-1}) = \varphi(a)\varphi(b)\varphi(a)^{-1}\varphi(b)^{-1} = e$ since it is abelian thus $A'_4 \in \ker(A_4 / \langle (12)(34), (13)(24) \rangle)$ and the kernel of this is $\langle (12)(34), (13)(24) \rangle$ thus $A'_4 \leq \langle (12)(34), (13)(24) \rangle$
 Consider $(123)(12)(34)(132)(12)(34) = (13)(24)$, $(124)(12)(34)(142)(12)(34) = (14)(23)$, and $(13)(24)(14)(23) = (12)(34)$ thus we know $A'_4 = \langle (12)(34), (13)(24) \rangle$ and since $\langle (12)(34), (13)(24) \rangle$ is abelian $\langle (12)(34), (13)(24) \rangle' = \{e\}$
 Thus it is solvable and the derived series is $A_4 \supseteq \langle (12)(34), (13)(24) \rangle \supseteq \{e\}$
 For C_{12} since it is abelian its derived series is $C_{12} \supseteq \{e\}$ thus it is solvable
 For D_6 we know it has a normal subgroup $\langle x^2 \rangle$ thus we know $D_6 / \langle x^2 \rangle$ is a group of order 4 which we know is abelian thus from similar proof to above we know $D'_6 \leq \langle x^2 \rangle$
 Consider $yx^2y^{-1}x^4 = x^8 = x^2$ thus $D'_6 = \langle x^2 \rangle$ and since $\langle x^2 \rangle$ is abelian its commutator subgroup is $\{e\}$ thus it is solvable
 $D_6 \supseteq \langle x^2 \rangle \supseteq \{e\}$
 For $C_6 \times C_2$ it is abelian thus $C_6 \times C_2 \supseteq \{e\}$ and it is solvable
 For $C_3 \rtimes C_4$ we know it has a normal subgroup C_3 thus $(C_3 \rtimes C_4) \setminus C_3$ is order 4 and thus abelian thus $(C_3 \rtimes C_4)' \leq C_3$
 Consider $bab^{-1}a^2 = a^4 = a$ thus $C_3 = (C_3 \rtimes C_4)'$ and since C_3 is abelian, $C'_3 = \{e\}$
 Thus it is solvable with $C_3 \rtimes C_4 \supseteq C_3 \supseteq \{e\}$
- (b)
- For $n = 1$, $S_1 = \{e\}$ thus done, and it is solvable
 - For $n = 2$, $S_2 \cong C_2$ thus it is abelian thus $S_2 \supseteq \{e\}$, and thus solvable
 - For $n = 3$, $S_3 \cong D_3$ thus it has a normal subgroup C_3 and since $D_3 / C_3 \cong C_2$ it is abelian thus $D'_3 \leq C_3$
 Consider $xyx^{-1}x^2 = x^4 = x$ thus $C_3 \leq D'_3$ thus $D'_3 = C_3$ thus since it is abelian $C'_3 = \{e\}$ thus it is solvable with $S_3 = D_3 \supseteq C_3 \supseteq \{e\}$
 - For $n = 4$, we know S_4 has normal subgroup A_4 thus $S_4 / A_4 \cong C_2$ thus abelian thus $S'_4 \leq A_4$
 Consider $(14)(123)(14)(132) = (142)$ thus since S'_4 is normal we can conjugate this 3-Cycle to get all 3-Cycles and thus $S'_4 = A_4$ and for A_4 look above thus it is solvable
 $S_4 \supseteq A_4 \supseteq \dots$
 - For $n \geq 5$ by similar proof above we can show that $S'_n = A_n$ and since A_n is simple we know that $A'_n = \{e\}$ or A_n if it is $\{e\}$ this would mean it is abelian which is not true thus it is A_n thus the derived series is $S_n \supseteq A_n \supseteq A_n \dots$ thus not solvable